

Week 11 - Weather and Forecasting: Worked Answers

EART23001 Atmospheric Physics and Weather

Workload guide. The shaded question headings show the intended workload:

- **IN CLASS:** selected anchor problems that we may work through together in the timetabled session.
- **YOU SHOULD TRY:** expected independent practice after the session.
- **EXTRA CHALLENGE:** optional problems for students who want additional depth or a harder application.

The categories are repeated here so the worked answers map directly onto the practice sheet. Use the solutions to check method, units and physical interpretation rather than only the final number. A course-wide Symbols, Constants and Units sheet is provided separately; non-standard symbols are also defined locally.

YOU SHOULD TRY Question 1: Identify the air mass from its trajectory

For each trajectory reaching the British Isles, identify the most likely air mass and predict one characteristic weather feature after modification along the path.

1. Air travels northeast from the subtropical North Atlantic near the Azores, remaining over ocean for several days.
2. Air arrives from Greenland and the high-latitude North Atlantic after a long ocean crossing.
3. Air moves north from North Africa across France during summer.
4. In winter, very cold continental air moves west from central/eastern Europe, then crosses the North Sea before reaching eastern Britain.

Worked answer

1. Tropical maritime: mild/warm and moist; often low cloud, drizzle and hill fog in exposed western areas.
2. Polar maritime: cool, moist and often unstable after ocean modification; showers with good visibility between showers are common.
3. Tropical continental: hot/warm and relatively dry, often hazy from dust/aerosol; thunderstorms can occur if moisture/instability increase.
4. Polar continental: cold and initially dry, but the North Sea crossing adds heat and moisture; eastern Britain can therefore receive snow showers in winter.

YOU SHOULD TRY Question 2: Identify the front from the observed change

Name the frontal type represented by each idealised description.

1. Surface air changes from 5°C to 11°C as a broad zone of layered cloud and steady rain passes.
2. Surface air changes from 12°C to 6°C as a narrow band of heavy showers and gusty winds passes.
3. A strong temperature contrast remains almost stationary over the same region for 18 hours, with persistent cloud and rain.
4. A faster cold front catches a warm front so that the warm sector is lifted away from the surface.

For cases 1 and 2, state whether the warmer air is replacing colder air or vice versa.

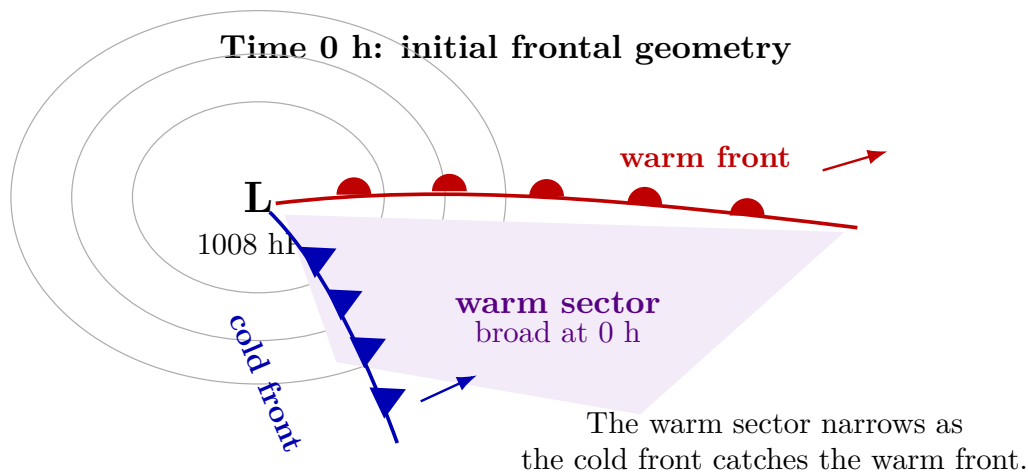
Worked answer

The fronts are: (1) warm front, with warmer air replacing colder surface air; (2) cold front, with colder air replacing warmer surface air; (3) stationary front; and (4) occluded front. The cloud/precipitation character in the first two cases is consistent with the gentler ascent ahead of a warm front and the steeper, more concentrated ascent along many cold fronts.

EXTRA CHALLENGE Question 3: Use pressure evolution to identify cyclone development and decay

An idealised mid-latitude cyclone has the following central pressure and surface warm-sector width:

Time (h)	0	12	24	36	48
Central pressure (hPa)	1008	996	984	988	997
Warm-sector width (km)	600	450	250	0	0



1. During which interval is the cyclone deepening most clearly?
2. When is the cyclone strongest according to central pressure?
3. At what time has surface occlusion clearly occurred according to the table?
4. What evidence indicates that the cyclone is filling/decaying after its mature stage?

Worked answer

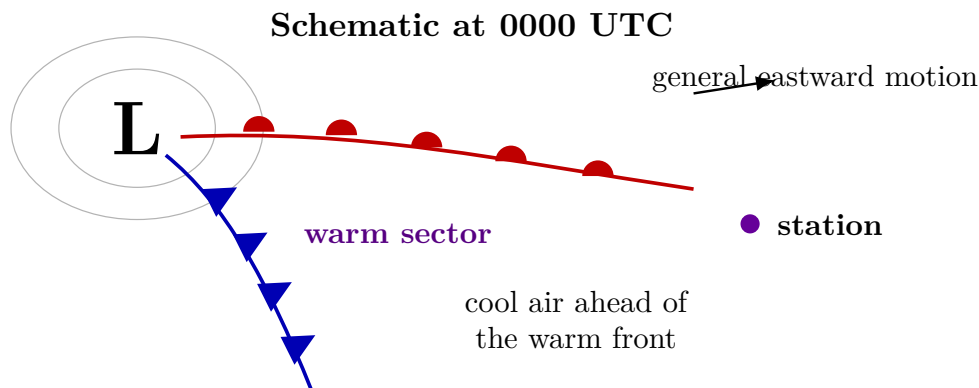
The cyclone deepens from 0 to 24 h as central pressure falls from 1008 to 984 hPa. It is strongest at about 24 h, when central pressure is lowest. By 36 h the surface warm sector has disappeared, indicating that occlusion has occurred. After 24 h the central pressure rises from 984 to 988 and then 997 hPa, so the low is filling and weakening while the warm sector remains absent.

IN CLASS Question 4: Diagnose a frontal passage from station observations

A station records the following sequence:

Time (UTC)	p (hPa)	T ($^{\circ}\text{C}$)	Wind	Cloud/weather
0000	1008	5	ESE	high cloud increasing
0300	1004	6	SE	cloud thickening
0600	999	8	S	steady rain
0900	996	11	SW	rain easing; low cloud
1200	994	12	SW	mild, cloudy, drizzle
1500	997	8	W	brief heavy showers, gusty
1800	1003	6	NW	showers clearing

The schematic below shows the broad synoptic setting at 0000 UTC. It is not intended to give the frontal passage times directly; use the station observations above to diagnose those.



1. At approximately what time did the warm front pass?
2. At approximately what time did the cold front pass?
3. For each front, quote at least three pieces of observational evidence from the table.
4. During which interval was the station most clearly in the warm sector?

Worked answer

The warm front most likely passes around 0900 UTC. Evidence includes the preceding steady rain and falling pressure, a marked temperature rise, a wind veer to southwesterly and rain easing as the station enters the warm sector.

The cold front most likely passes around 1500 UTC. Evidence includes the temperature fall from 12 to 8°C, a wind veer from southwest to west, pressure beginning to rise and a short period of heavy/gusty showers. By 1800 the wind is northwesterly and showers are clearing.

The station is most clearly in the warm sector from roughly 0900 to just before 1500 UTC.

EXTRA CHALLENGE Question 5: Use isobar spacing on a synoptic chart

A surface analysis has 4 hPa isobar intervals. Near Region A, adjacent isobars are typically 80 km apart; near Region B they are 200 km apart. Assume similar latitude and air density.

1. Which region has the larger horizontal pressure gradient?
2. By what factor is it larger?
3. If the upper-air geostrophic wind in Region B is about 12 m s^{-1} , estimate the corresponding value in Region A under the same assumptions.
4. Why should a forecaster not expect the observed 10-m wind to equal this geostrophic estimate exactly?

Worked answer

Region A has the larger pressure gradient because the same pressure change occurs across a shorter distance. The ratio is

$$\frac{|\nabla p|_A}{|\nabla p|_B} = \frac{200}{80} = 2.5.$$

At similar latitude and density, geostrophic wind is proportional to the pressure gradient, so

$$v_{g,A} \approx 2.5(12) = 30 \text{ m s}^{-1}.$$

Near the surface, friction slows the wind and makes it cross isobars toward lower pressure. Terrain, buildings, stability and curvature also make real 10-m winds depart from the ideal geostrophic value.

YOU SHOULD TRY Question 6: Put the NWP forecast cycle in order

The following stages are deliberately scrambled:

- compare the forecast with later observations;
- collect surface, radiosonde, aircraft, radar and satellite observations;
- numerically integrate the atmospheric governing equations forward in time;
- combine observations with a previous forecast using data assimilation;
- produce the best estimate of the three-dimensional initial atmospheric state;
- represent unresolved processes such as some cloud/turbulence/surface exchange using parameterisations.

Put the stages into a sensible forecast-cycle order. Then identify two distinct sources of forecast error.

Worked answer

A sensible order is:

1. collect surface, radiosonde, aircraft, radar and satellite observations;
2. combine observations with a previous forecast using data assimilation;
3. produce the best estimate of the three-dimensional initial atmospheric state;
4. numerically integrate the atmospheric governing equations forward in time;
5. represent unresolved processes such as some cloud/turbulence/surface exchange using parameterisations;
6. compare the forecast with later observations.

In an operational model, the parameterisations in step 5 are applied during the numerical integration in step 4; they are written separately here only so that the stages in the question map directly onto the worked answer.

Two important error sources are uncertainty/incompleteness in the initial atmospheric state and imperfections in the model, including finite resolution and parameterised unresolved processes. Numerical approximation and chaotic growth of small errors also contribute.

YOU SHOULD TRY Question 7: Interpret an ensemble forecast probabilistically

A 20-member ensemble forecast for Manchester gives the following outcomes for tomorrow afternoon:

- 14 members produce measurable rain;
 - 6 members produce more than 10 mm of rain;
 - 4 members remain completely dry.
1. Estimate the ensemble probability of measurable rain.
 2. Estimate the ensemble probability of more than 10 mm.
 3. Estimate the probability of a completely dry outcome.
 4. A single deterministic run predicts 12 mm. Explain why it would be misleading to report “12 mm will fall” without the ensemble information.

Worked answer

The probabilities estimated from the ensemble are

$$P(\text{rain}) = \frac{14}{20} = 70\%, \quad P(> 10 \text{ mm}) = \frac{6}{20} = 30\%, \quad P(\text{dry}) = \frac{4}{20} = 20\%.$$

The deterministic 12 mm solution is only one plausible trajectory. The ensemble shows substantial uncertainty: some members are dry and only 30% exceed 10 mm. The forecast should therefore communicate probability/range rather than presenting one run as certain.

EXTRA CHALLENGE Question 8: Diagnose a blocking-high scenario

For four consecutive days, analysis charts show a 1035 hPa anticyclone remaining almost stationary over western Europe. Atlantic cyclones that would normally move east are instead diverted north around it.

1. Why is this described as a blocking high?
2. What broad surface-wind and vertical-motion pattern is expected near the centre of the anticyclone?
3. Give one likely summer weather consequence beneath the block and one possible winter consequence.
4. Why can regions around the edge of a block experience persistent wet weather even while the centre is dry?

Worked answer

It is a block because the large, persistent high-pressure system disrupts the normal eastward progression of mid-latitude cyclones. Near its centre, surface flow is weak and divergent with broad subsidence aloft. In summer this can favour prolonged dry, sunny and potentially hot conditions. In winter, weak winds and subsidence inversions can favour frost, fog and poor air quality.

Cyclones and their fronts can be slowed or repeatedly steered around the flanks of the block, so a neighbouring region may remain beneath a persistent storm track or frontal zone and receive repeated precipitation.

EXTRA CHALLENGE Question 9: AI and physics-based forecasting

A conventional NWP model and a data-driven machine-learning model both produce a 5-day forecast. The ML model runs much faster, while the NWP model explicitly evolves conservation laws and parameterised physical processes.

1. Give one scientific or operational advantage of the ML approach.
2. Give one advantage of retaining physics-based NWP.
3. Explain why a forecasting centre might use both rather than regard them as mutually exclusive replacements.

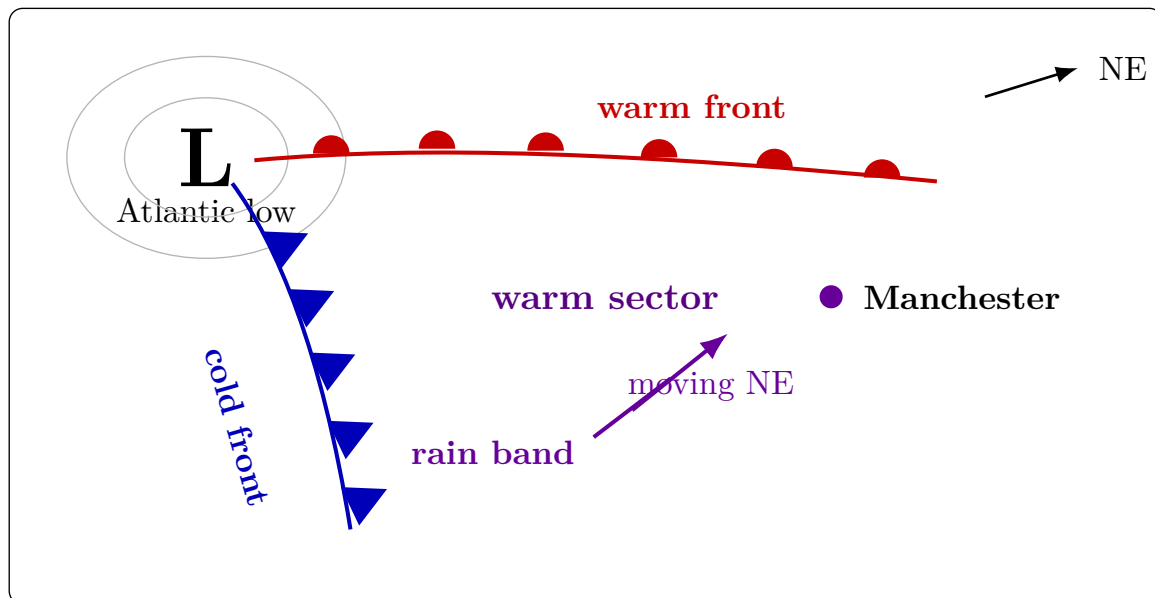
Worked answer

A trained ML model can generate forecasts very rapidly and can learn useful high-dimensional relationships from large reanalysis/observation datasets. Physics-based NWP retains an explicit dynamical/thermodynamic structure, supports process diagnosis and remains closely integrated with data assimilation and physically based specialist prediction.

Using both can exploit complementary strengths: fast ML forecasts can add ensemble members or guidance, while physics-based models provide physically interpretable predictions and a framework for processes/conditions that may be poorly represented in training data. Operational systems can compare, blend and verify both types of guidance.

IN CLASS Question 10: Integrative UK weather case: chart, sounding and radar

At 0900 UTC a deep Atlantic low lies west of Ireland. A warm front is moving northeast across Wales and a cold front is approaching southwest England. Manchester lies in the warm sector. Surface observations at Manchester are $T = 11^\circ\text{C}$, dew point 9°C , pressure 995 hPa falling at 2 hPa in 3 h, and wind 210° at 12 m s^{-1} . A nearby sounding shows a nearly saturated layer from 1–3 km, a freezing level near 2.2 km and winds increasing to 30 m s^{-1} by 5 km. Radar shows a narrow band of heavier precipitation 120 km to the southwest moving northeast at 50 km h^{-1} .

Schematic synoptic situation at 0900 UTC

1. Estimate when the radar rain band will reach Manchester if its speed remains constant.
2. Use the surface temperature/dew-point pair and sounding to assess whether cloud is likely before frontal passage.
3. Explain why the falling surface pressure and strengthening winds aloft are dynamically consistent with an approaching cyclone/frontal system.
4. What precipitation phase is most likely at the surface, and why?
5. Predict the likely changes in wind, temperature and pressure as the cold front passes.
6. Identify one important uncertainty in this six-hour forecast that cannot be removed by the information supplied.

Worked answer

At 50 km h^{-1} , a band 120 km away would arrive in about

$$t = \frac{120}{50} = 2.4\text{ h},$$

so roughly 2–3 hours later if motion and structure are maintained.

At the surface, the dew-point depression is only 2 K, and the sounding is nearly saturated through 1–3 km. Low/mid-level cloud is therefore very likely. The falling pressure is consistent with the approach/deepening influence of a low-pressure system. Stronger winds aloft are also typical of frontal zones and the stronger horizontal pressure gradients associated with synoptic-scale cyclones.

The freezing level is around 2.2 km, so hydrometeors generated as ice aloft have a substantial above-freezing layer in which to melt. Rain is therefore the most likely surface precipitation phase.

As the cold front passes, the wind is likely to veer, temperature to fall, pressure to stop falling and then rise, and a short period of heavier rain or showers may accompany the frontal zone. Behind the front, cooler air and more broken/showery cloud are plausible.

Uncertainty remains in the exact frontal/radar-band speed and intensity, small-scale convective enhancement, and the evolution of the cyclone itself. The supplied observations provide a diagnosis, not a deterministic guarantee of the next six hours.